Project 4 Report

Kathryn Mew

ST: CMSC 395 Android Programming

May 1, 2024

- **Project Title:** KomosForge – Event Planner App

- **Team Members and Individual Responsibilities:**

I worked independently on this project, and held all responsibilities

- **Overall Goal of the App:**

Utilize Calendar functionalities and connection abilities to provide a user-friendly event scheduler where users can connect and plan events while minimizing back-and-forth conversation about date and times. One of the key features of this app is that it allows users to see the time periods for which their friends are occupied for any given day without sharing details of these personal events!

- **Functionality:** Explain the purpose of various parts of the app.

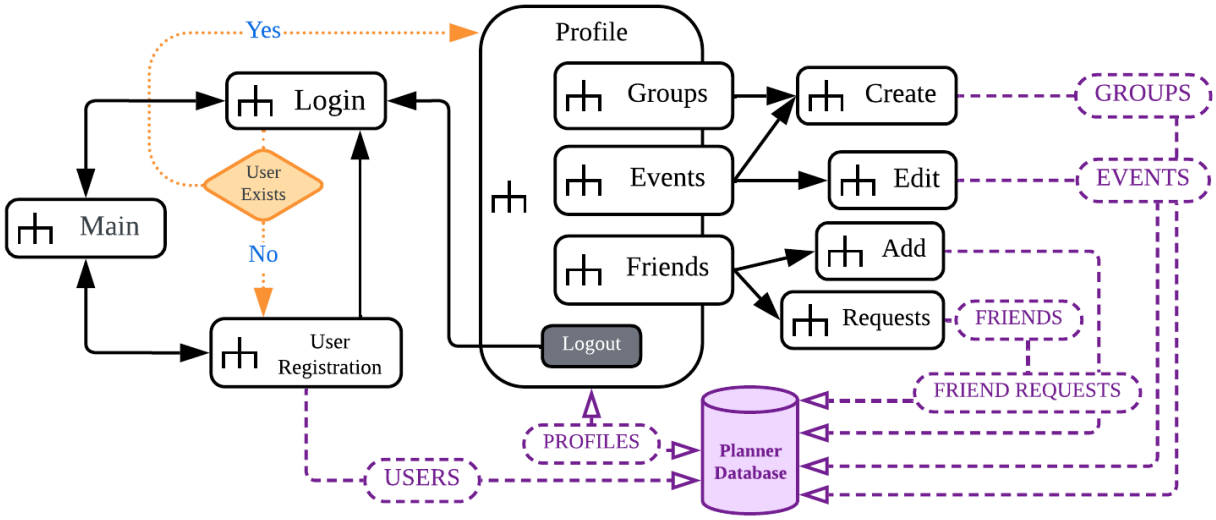


Figure 1: Flow Diagram with Database Interaction

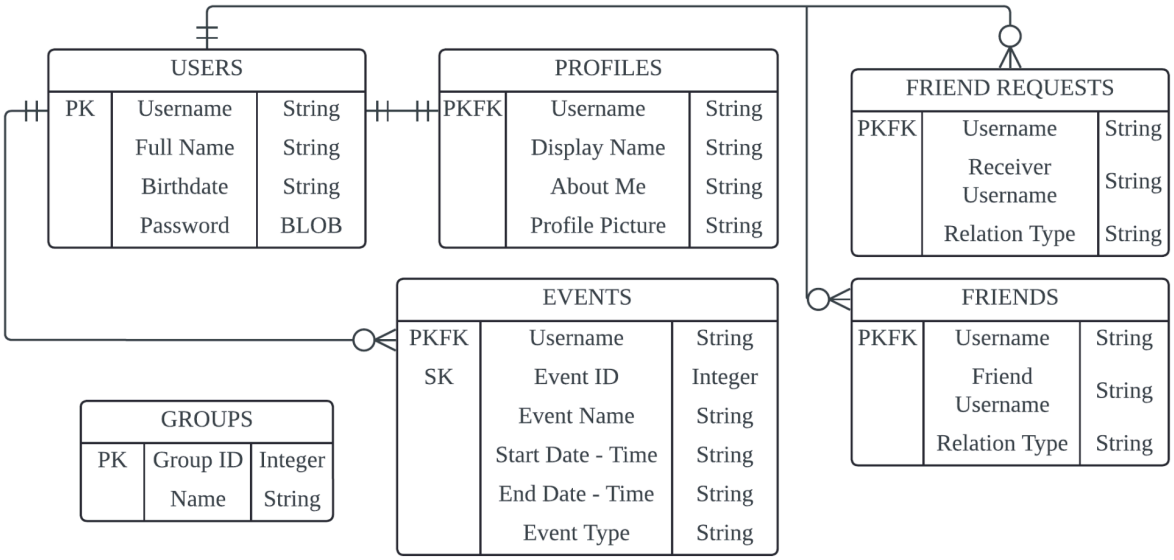


Figure 2: Planner Database

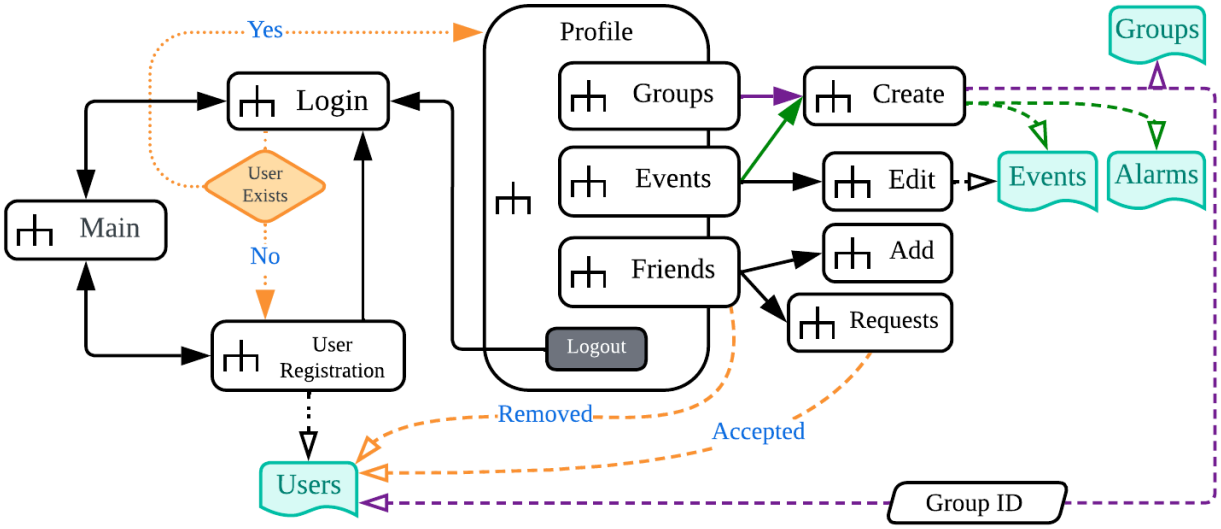
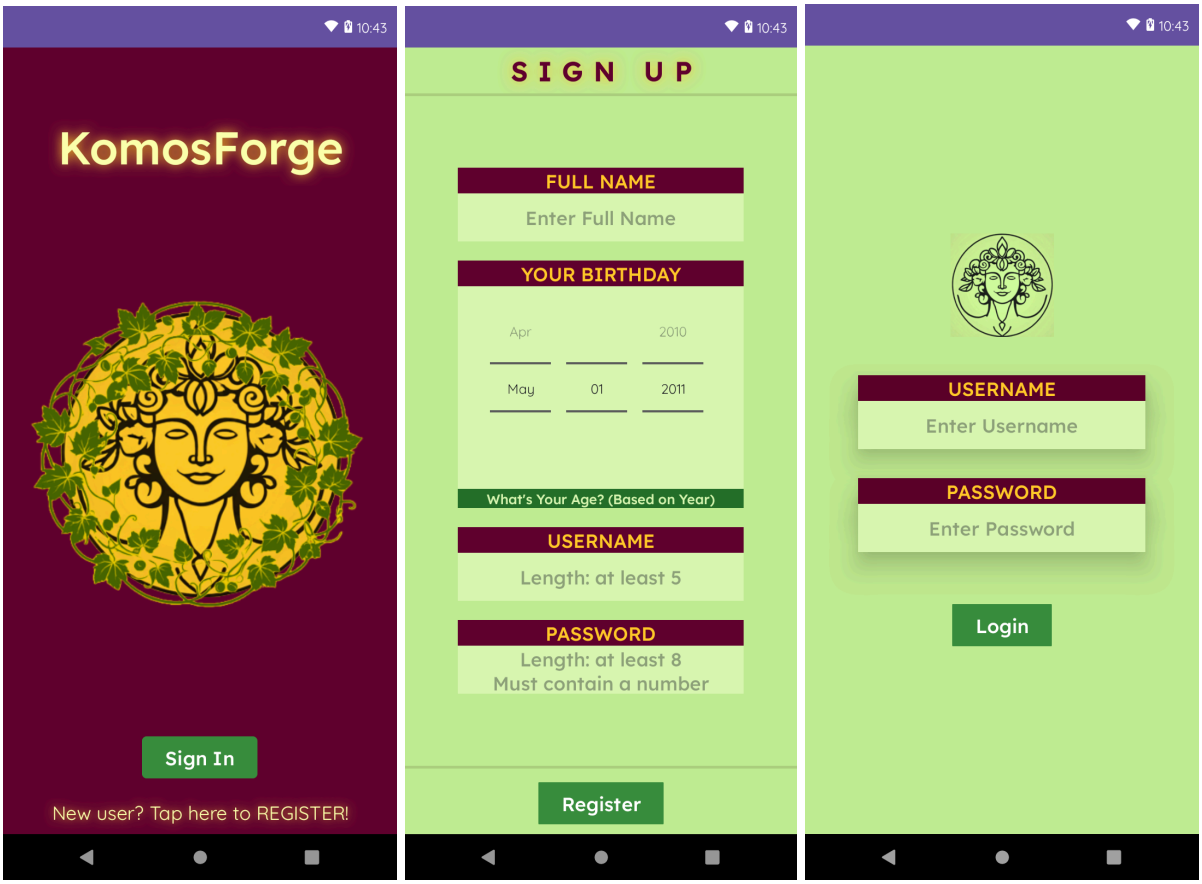
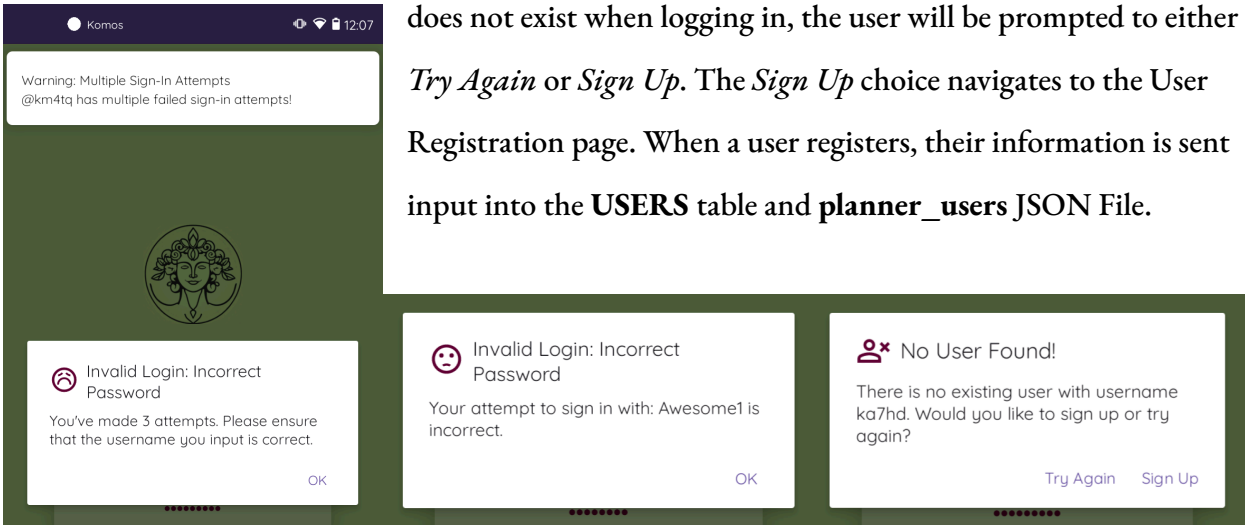
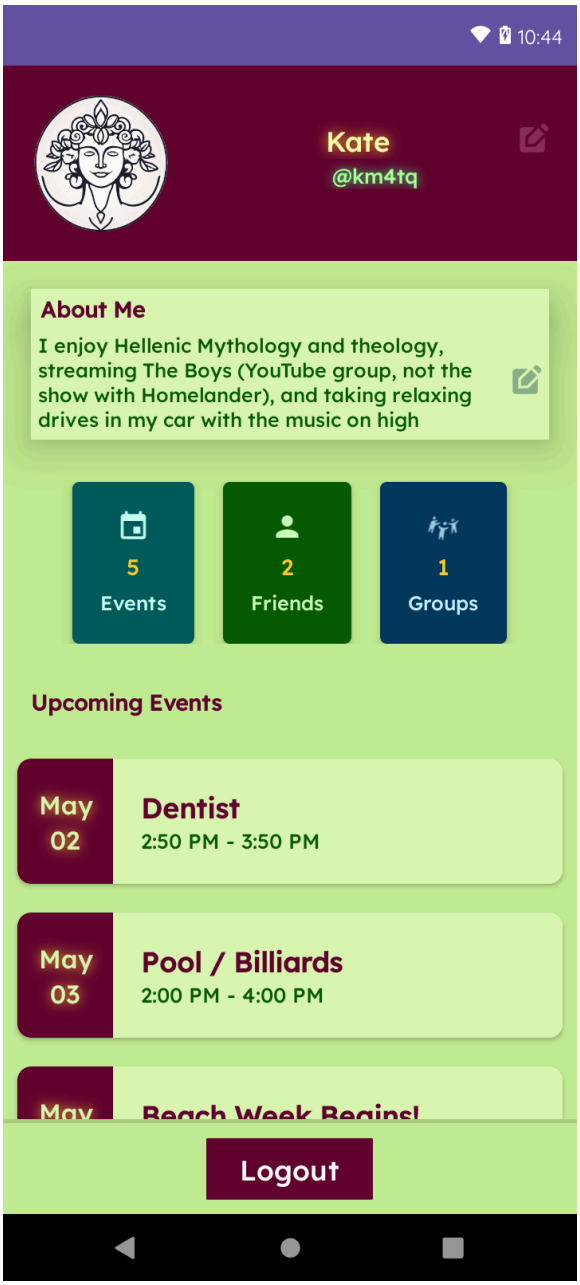


Figure 3: Flow Diagram with JSON Files



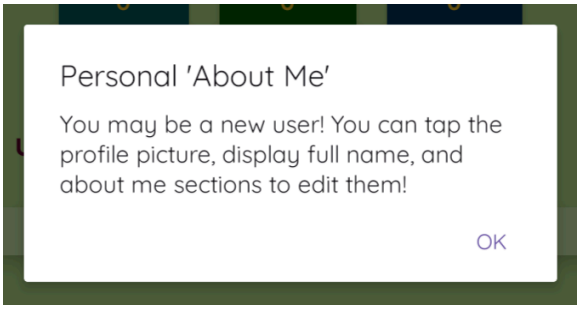
These three screens are the entry gates into the application. The registration and sign-in pages utilize AlertDialogs when certain conditions are met, including non-existent usernames/users, incorrect passwords and failed attempts. For the registration page, if a username already exists, regardless of individual letter casing, then the new user will not be able to register. If a username does not exist when logging in, the user will be prompted to either *Try Again* or *Sign Up*. The *Sign Up* choice navigates to the User Registration page. When a user registers, their information is sent input into the **USERS** table and **planner_users** JSON File.



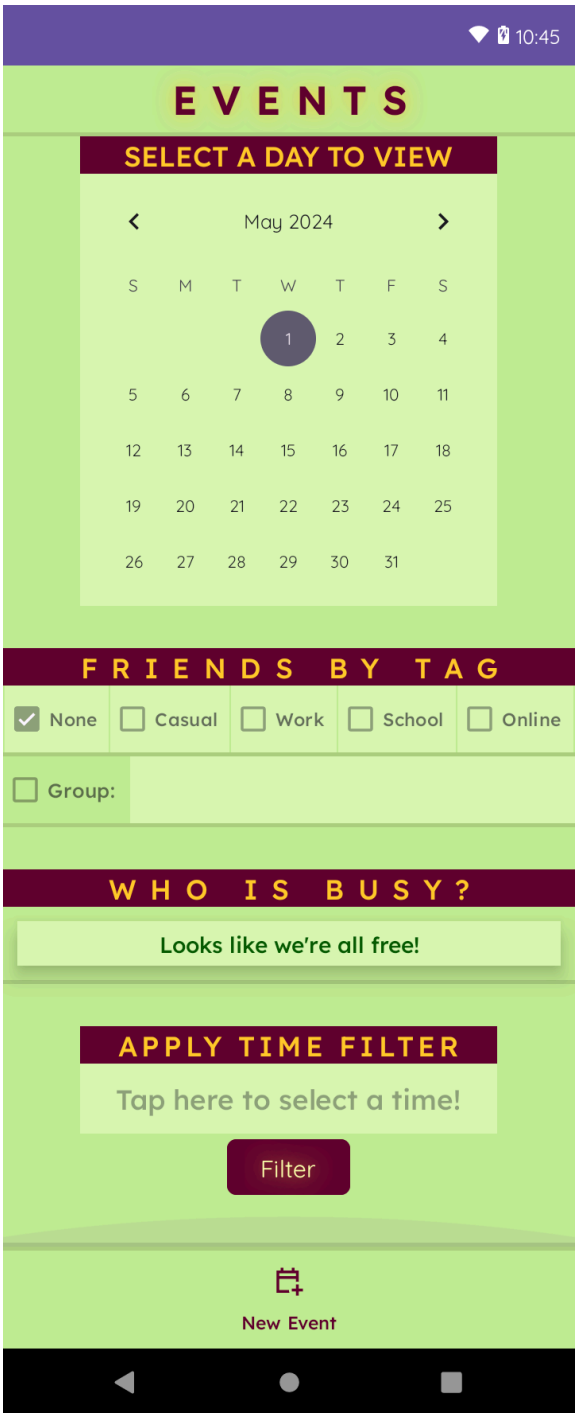


Profile Activity:

This is the user's "homepage". When a new user first logs in, they will be welcomed with:



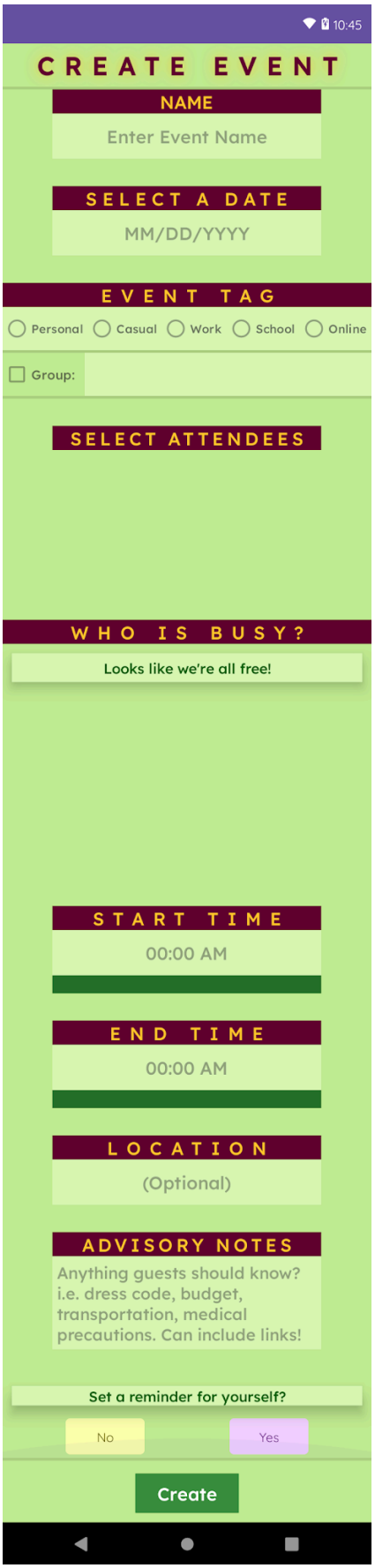
As the user creates new events, the *Upcoming Events* section may be updated if there are any events scheduled within a week-period of time from the current day. The numbers on the navigation buttons to the **Events**, **Friends**, and **Groups** activities are updated as changes are made to the respective categories. As of now, the profile picture is not modifiable, but that will hopefully be implemented in the future!



Events Activity:

The **Events** activity is where the key feature of this application is implemented! Due to some coding technicalities, in order for results to be queried, the user must select a day to view, even if the day is the current date. When this occurs, the checkboxes in the *Friends by Tag* section will be enabled and clickable. The default results shown in the *Who is Busy?* section are only the user's events. If None or Group is checked, then no other buttons should be. Additionally, when the Group box is checked, the spinner beside the checkbox will be enabled and allow for the user to choose which group of their friends they'd like to see the availability of.

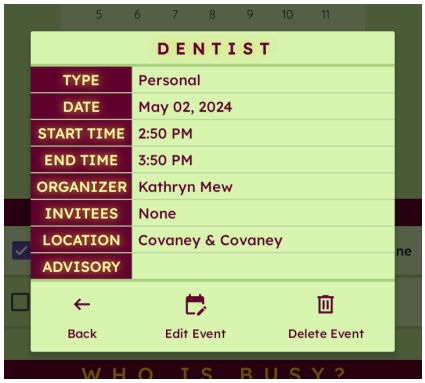
Just as the widget mentioned before, the *Apply Time Filter* TextView is only accessible when a date is selected. Any time picked by the user is applied only when the Filter button is clicked. From this activity, the user can do three actions: create new events, view event details, and edit existing events. The view event details and edit existing events actions are only available when a particular event appears in the *Who is Busy?* section.



Create Event Activity:

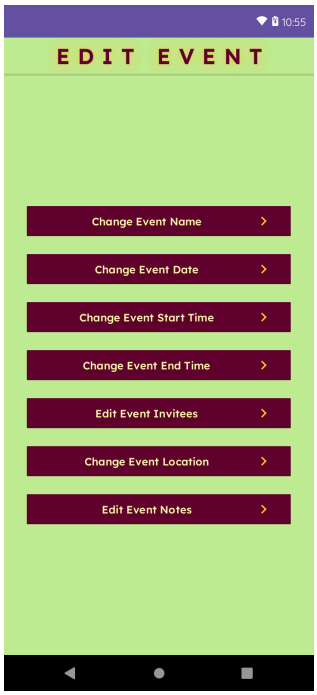
This activity uses the same idea from its parent activity; it provides users the ability to see their friends' availability, only this time they are able to use this information while scheduling an event at the same time. Just as the **Events** activity, no other fields will be available for user interaction or accept input until a date is selected. The *Select Attendees* recycler is populated by a checkbox list when an event tag other than "Personal" is selected. The *Alarm* configuration is not available until a date and times are selected.

View Event (Button Press):

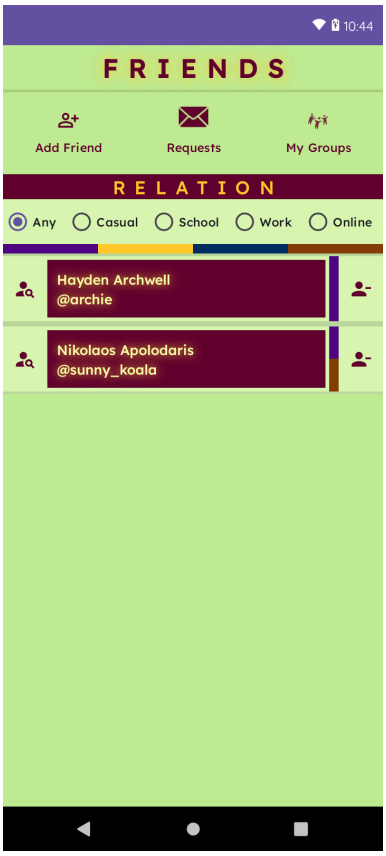


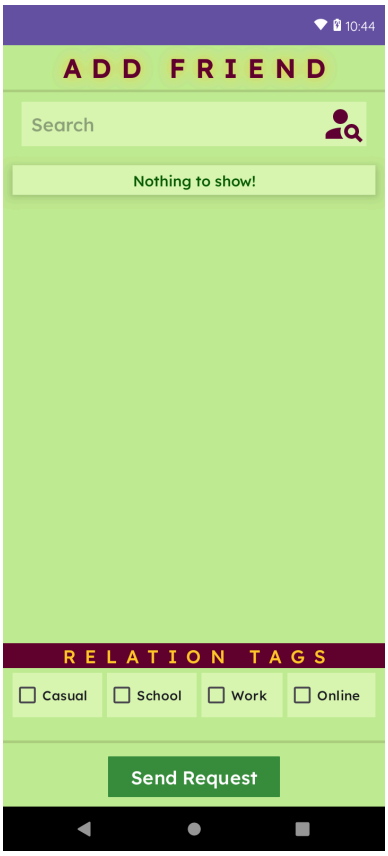
Accessed from the **Events** activity. This feature is only available if the event is personal for the logged in user or a shared event with the user. From here, the event can either be deleted or edited.

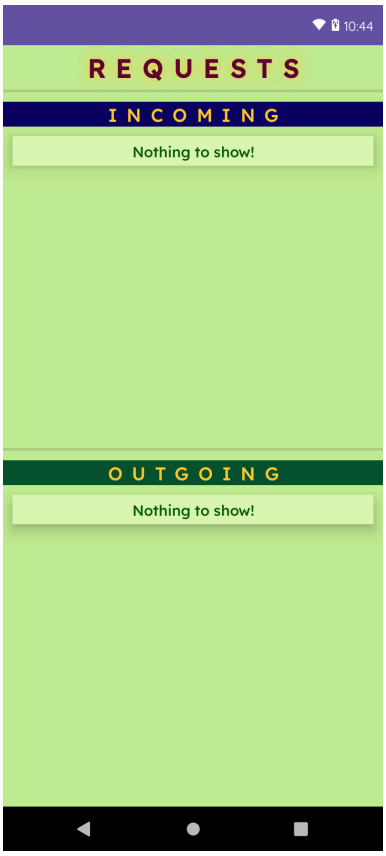
Edit Events Activity:



Utilizes AlertDialogs with EditTexts to allow for users to see what the current value is before changing it. Changes are not finalized until the user presses "Confirm" on the dialog.





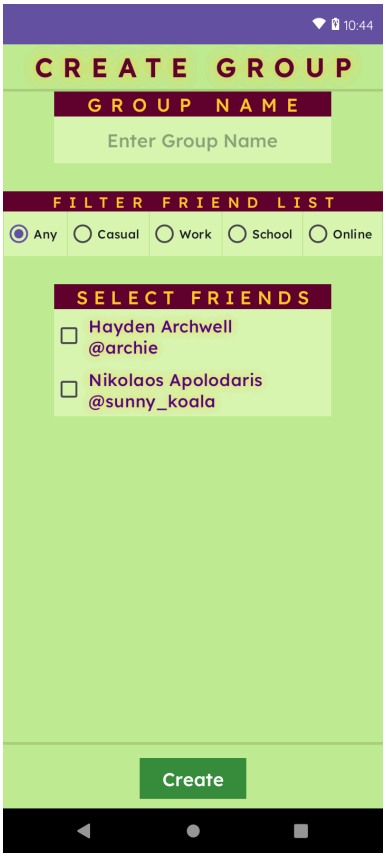


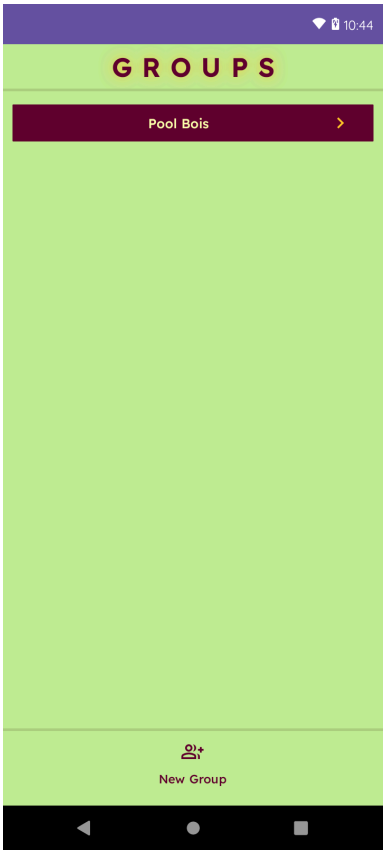
Friends Activity:

Utilizes RadioButtons, and a RecyclerView with TextViews and ImageButtons to display the user's friends. From here, the user can access **Add Friends**, **Requests**, and **My Groups** (subject to change).

Add Friends Activity:

A user can search for friends by entering either a part or all of another user's full name, display name, or username. Current search results show any result that has some substring of the user-input keyword, but I would like to improve this one day. Only one person can be added at a time. Then, the user can select one or more relation tags to categorize their friends. When a request is sent, it appears as outgoing for the user in **Friend Requests**, and incoming for the user that the request was sent to.





Groups Activity:

Uses a recycler view to list any and all groups a user is a member of. No further action can be done if a group is selected, but it would navigate to a page showing all the members of the group with options to add or remove members, edit group name, or delete the group. From here, the user can create a new group,

Create Group Activity:

A group must consist of two or more members, including the creator. For now, there is no specific owner of the group documented, which means any member would be able to edit or delete the group if that functionality was provided.

- Components included (List of widgets involved):
  - TextView & EditText
  - Buttons, ImageButtons, Checkboxes, RadioGroup/RadioButtons
  - Spinner, TimePicker, NumberPicker, and DatePicker & CalendarView

- **Specifications**:

Username - Length of at least 5, can be alphanumeric. Cannot be an existing username; the existsUsername() method is not case-sensitive. i.e. KM4tq is the same as km4tq, and will not be a valid username to sign up with if there already exists a user with this name.

Password - Length of at least 8 and contain at least one number.

Birth Date - Time Picker is already set so that any person under 13 cannot sign up.

- **Overall findings while working on this app**:

I enjoyed being able to experiment with CalendarView and TimePickers, with other Fragment-type dialogs like the DateDialog. I learned how to work with JSON Files and SQLiteDatabase operations. I was able to advance my knowledge in using RecyclerViews. A few tough lessons that I took from this project while working through are: time management, idea → implementation restraint, and that waiting until the last minute to debug often hides sneaky bugs.

- **Future directions and Improvements**:

I would like to fix any incomplete or incorrectly operating functionalities before I implement any new features. These functionalities in need of fixing or implementation and my plan to repair the issues, by activity, are:

  - Add Friends: Toast message displays the first result in the list containing which users to show to the user.
  - Create Event: Users cannot select days before a selected date if the previous date selected is saved into the same Calendar instance used to verify if the selected date is after the current date. My solution would be to create a separate Calendar instance to store the user-chosen date.

- Events: Time filter should show any events at or after the desired filter time exclusively on the same day.

Further implementations / expansions to add one day:

- Alarms: Background service & scheduling alarms made for events by different users
- Events:
    - Implement adding and removing invitees when editing events
    - Implement organizer permission for authorized invitees to edit shared events
    - Implement uploading documents in advisory notes section, and clickable links in location
    - Improve the categorization of events → event types will not be the same as relation types.
    - (Future Installment) Change the CalendarView selection in Events to a clickable TextView displaying the current day, with navigation arrows on either side. Users will have two ways to change the date.
- Friends: Improve searching for people in Add Friends to provide concurrent search results
- Groups:
    - Implement viewing group members
    - (Future installment) The Groups activity acts as a directory for groups a user is in. When a group is selected, navigate to the corresponding group chat.
- Profiles: Implement uploading and storing profile pictures

If I were to expand the uses of this app, I would include a messenger service for one-to-one and group chats. This way, users would not have to exit the application to communicate with other messenger applications. Another idea would be to provide users a way to store photos taken related to the events they create; a yearbook of sorts.

I enjoyed working on this project, as well as the course overall. It was a fun course that allowed for me to tap into my creativity, which is something I've not been able to do for a while!